Python Activity-Based Learning
Program 4: Employee Payroll and Deduction Analysis

Problem Statement

Calculate employee gross salary using basic pay, allowances, overtime, and incentives. Apply progressive deductions and generate a salary statement.

Objectives

• Calculate allowances and overtime pay.

• Calculate gross salary and progressive deductions.

• Display net salary in a formatted statement.

Assumptions

• HRA is 20% and other allowance is 10% of basic pay.

• Overtime is paid at 1.5 times the hourly basic rate, using 26 working days and 8 hours per day.

• Deduction: 5% on the first ■20,000 of gross salary and 10% on the remaining amount. These are illustrative assignment rules.

Algorithm

1. Read employee name, basic pay, overtime hours, and incentive.

2. Validate non-negative values.

3. Calculate HRA, other allowance, and overtime pay.

4. Calculate gross salary.

5. Calculate progressive deduction.

6. Calculate net salary and display the salary statement.

Flowchart

```
START → Read employee data → Validate values → Calculate allowances and overtime → Calculate gross salary → Ca
```

Pseudocode

```
BEGIN
  READ name, basic, overtime_hours, incentive
  IF any value < 0 THEN DISPLAY invalid input
  hra = 20% of basic
  other = 10% of basic
  hourly_rate = basic / (26 * 8)
  overtime_pay = overtime_hours * hourly_rate * 1.5
  gross = basic + hra + other + overtime_pay + incentive
  IF gross <= 20000 THEN deduction = 5% of gross
  ELSE deduction = 1000 + 10% of (gross - 20000)
  net = gross - deduction
  DISPLAY salary statement
END
```

Python Source Code

```
name = input("Enter employee name: ")
basic = float(input("Enter basic pay: "))
overtime_hours = float(input("Enter overtime hours: "))
incentive = float(input("Enter incentive: "))

if basic < 0 or overtime_hours < 0 or incentive < 0:
    print("Invalid input. Values cannot be negative.")
else:
    hra = basic * 0.20
    other_allowance = basic * 0.10
    hourly_rate = basic / (26 * 8)
```

```
overtime_pay = overtime_hours * hourly_rate * 1.5

gross = (basic + hra + other_allowance
         + overtime_pay + incentive)

if gross <= 20000:
    deduction = gross * 0.05
else:
    deduction = 20000 * 0.05 + (gross - 20000) * 0.10

net_salary = gross - deduction

print("\nEMPLOYEE SALARY STATEMENT")
print("Employee:", name)
print("Basic pay: Rs.", round(basic, 2))
print("HRA: Rs.", round(hra, 2))
print("Other allowance: Rs.", round(other_allowance, 2))
print("Overtime pay: Rs.", round(overtime_pay, 2))
print("Incentive: Rs.", round(incentive, 2))
print("Gross salary: Rs.", round(gross, 2))
print("Deduction: Rs.", round(deduction, 2))
print("Net salary: Rs.", round(net_salary, 2))
```

Test Case

Name: Arun; basic pay ■20,000; overtime 8 hours; incentive ■1,000.
Expected: HRA ■4,000; other allowance ■2,000; overtime pay about ■144.23; gross about ■27,144.23; deduction about ■1,714.42; net about ■25,429.81.

Sample Output

```
EMPLOYEE SALARY STATEMENT
Employee: Arun
Basic pay: Rs. 20000.0
HRA: Rs. 4000.0
Other allowance: Rs. 2000.0
Overtime pay: Rs. 144.23
Incentive: Rs. 1000.0
Gross salary: Rs. 27144.23
Deduction: Rs. 1714.42
Net salary: Rs. 25429.81
```

Conclusion

The program prepares a salary statement by calculating allowances, overtime, gross salary, progressive deductions, and net salary.